

IAS PRELIMS/MAINS

Ecology & Environment

A stylized illustration of a green city skyline with various skyscrapers, wind turbines, and a sun, set against a light green background with a globe-like shape.

A COMPREHENSIVE COVERAGE

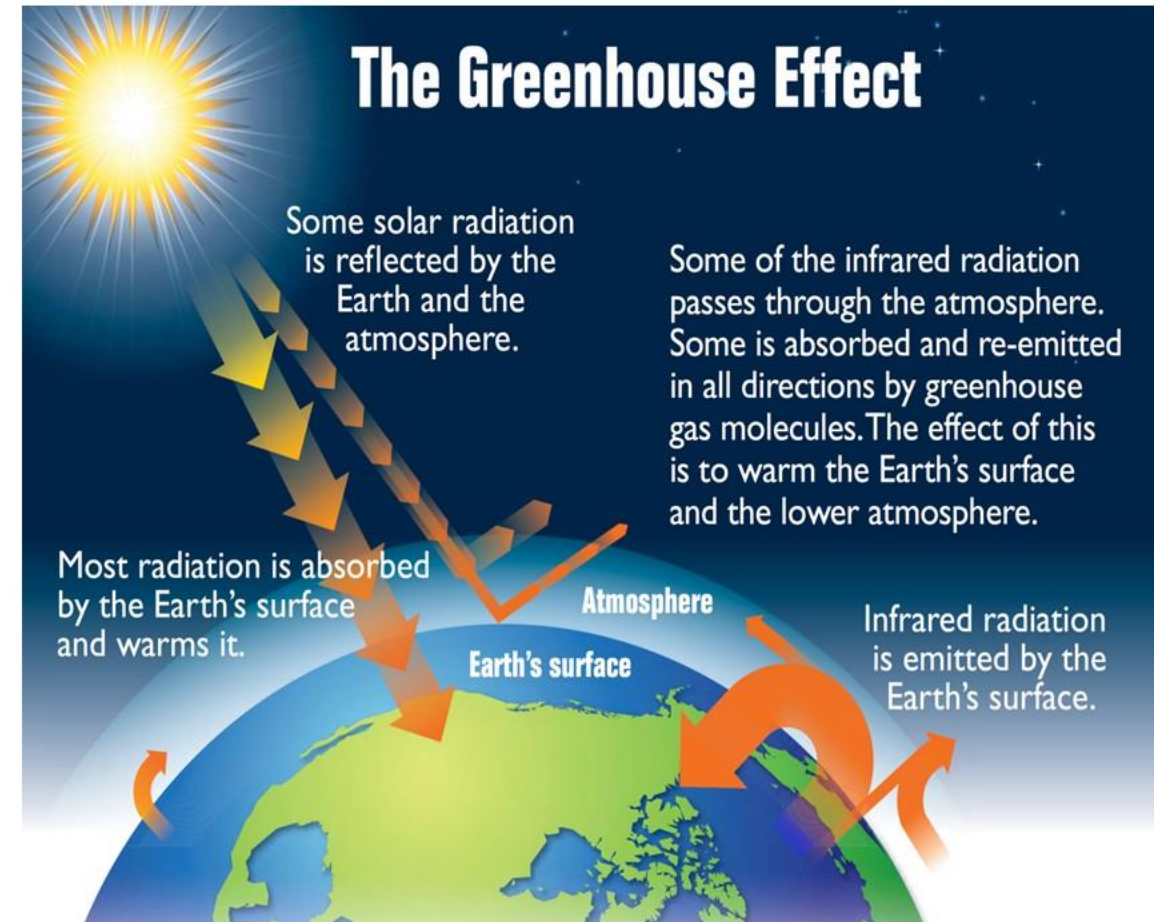
MODULE - 40

CLIMATE CHANGE - 1

CLIMATE CHANGE

GREENHOUSE EFFECT AND GLOBAL WARMING

- Greenhouse effect is the warming of Earth's surface and troposphere caused by the presence of Greenhouse Gases (water vapor, carbon dioxide, methane etc.).
- The atmosphere allows most of the solar radiation to pass through and reach Earth's surface. As Earth's surface is heated, it radiates some energy back towards the space.
- This radiation is absorbed by the greenhouse gases in the atmosphere (troposphere), raising its temperature. The heated atmosphere in turn emits radiation back towards the Earth's surface. This phenomenon is known as Greenhouse Effect.



CLIMATE CHANGE



GREENHOUSE EFFECT AND GLOBAL WARMING

- Global warming is the phenomenon of increasing average air temperatures near the surface of Earth over the past one to two centuries due to increasing concentrations of Greenhouse gases in the atmosphere.
- Global warming can alter the Earth's long term climatic conditions and produce new patterns of extreme conditions such as droughts and heavy rainfall.
- Climate Change - It refers to changes in weather patterns over a long period due to the growing accumulation of greenhouse gases in the atmosphere.



CLIMATE CHANGE

GREENHOUSE EFFECT AND GLOBAL WARMING

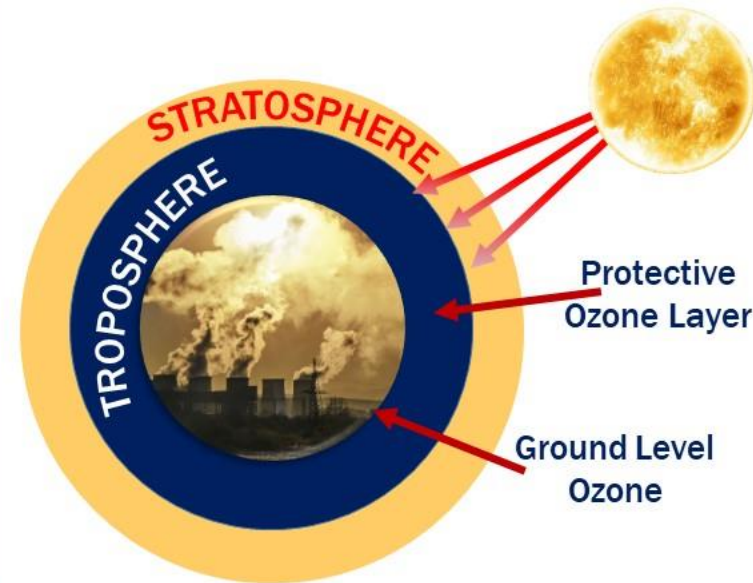
- Without the heating caused by the greenhouse effect, Earth's average surface temperature would be only about -18°C .
- On Venus the very high concentration of carbon dioxide and sulfur dioxide in the atmosphere causes an extreme greenhouse effect resulting in surface temperatures as high as 450°C .
- Although the greenhouse effect is a naturally occurring phenomenon, it is intensified by the emission of greenhouse gases into the atmosphere due to human activity.
- From the beginning of the Industrial Revolution to the present time, the amount of carbon dioxide in the atmosphere increased by roughly 30 to 40 percent and the amount of methane more than doubled.



CLIMATE CHANGE

GREENHOUSE GASES

1. **Water Vapor** - Is the largest greenhouse gas in the atmosphere in terms of volume. It is the biggest contributor to Greenhouse effect. Humans are not responsible for emitting this gas in quantities sufficient to increase its concentration in the atmosphere.
2. **Carbon Dioxide** - Second most abundant greenhouse gas in the atmosphere. It is the main Greenhouse gas emitted from burning of fossil fuels.
3. **Methane**
4. **Nitrous Oxide** - Is also an Ozone Depleting Substance.
5. **Sulfur Dioxide**
6. **Troposphere/Ground Level Ozone.**



CLIMATE CHANGE

FLUORINATED GASES

- Fluorinated gases are
 1. Chlorofluorocarbons (CFCs)
 2. Hydrofluorocarbons (HFCs)
 3. Perfluorocarbons (PFCs)
 4. Sulfur Hexafluoride
- Fluorinated gases have a high Global Warming Potential (GWP). It means they are the most potent and longest lasting type of greenhouse gases.



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